

CS4084

Quantum Computing

Week 07 – Lecture 13-14

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Outline

- 1 Quick Review: What We have Covered Last Lecture
- 2 Quantum Circuits
- 3 Inner Product
- 4 Orthogonality and orthonormality
- 5 Projections
- 6 Summary!

Quick Review: What We have Covered Last Lecture I

- **Probabilistic States:** Compound systems associate probabilities with the **Cartesian product** of individual state spaces.
- **System Independence:** Two systems are independent if

$$Pr((X, Y) = (a, b)) = Pr(X = a)Pr(Y = b)$$

- **Probabilistic Operations:** Transformations on compound systems can be represented using **stochastic matrices** acting on the combined state space.
- **Tensor Product:** The fundamental operation used to combine vectors of multiple systems:

$$|\pi\rangle = |\phi\rangle \otimes |\psi\rangle$$

Quick Review: What We have Covered Last Lecture II

- **Vector Representation:** Joint states follow **lexicographical ordering**, e.g.

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

- **Matrix Tensor Product:** Operations acting independently on subsystems are represented using the **Kronecker product** of matrices ($M \otimes N$).
- **Multi-Qubit Space:** A system of n qubits exists in the tensor product space

$$(\mathbb{C}^2)^{\otimes n}$$

- **Local vs Global Operations:**
 - Local operations act on individual qubits (e.g., $H \otimes I$).
 - Global operations (e.g., CNOT) operate on multiple qubits and enable **entanglement**.

Quick Review: What We have Covered Last Lecture III

- **Linearity Property:** The tensor product is **bilinear**:

$$A \otimes (B + C) = A \otimes B + A \otimes C$$

Quantum Circuits

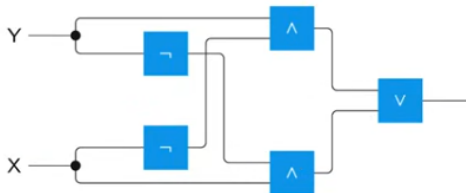
Circuits are models of computation:

- Wires carry information
- Gates represent operations

In this series, circuits are always **acyclic** — information flows from left to right.

Example: Boolean circuits

Wires store binary values, gates represent Boolean logic operations, such as AND ($\wedge$), OR ($\vee$), NOT ($\neg$), and FANOUT ($\bullet$).



Another Example of Arithmetic Operations in Quantum Circuits

In the **quantum circuit** model, the wires represent qubits and the gates represent both unitary operations and measurements.

Example



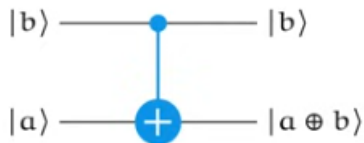
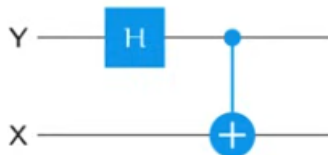
$$H = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad T = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{pmatrix}$$

$$THSH = \begin{pmatrix} \frac{1+i}{2} & \frac{1-i}{2} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{pmatrix}$$

Example: Multiple Figures

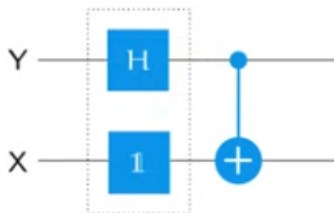
Convention

In this series (and in Qiskit),
ordering qubits from
bottom-to-top is equivalent
to ordering them *left-to-right*.



Example

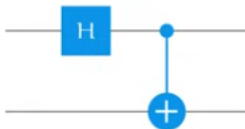
Example



$$\mathbf{1} \otimes H = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

Circuit action on a standard basis measurement!

Example



$$|00\rangle \mapsto |\phi^+\rangle$$

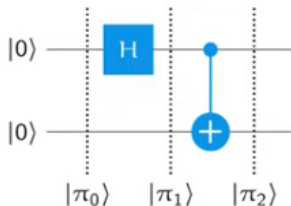
$$|01\rangle \mapsto |\phi^-\rangle$$

$$|10\rangle \mapsto |\psi^+\rangle$$

$$|11\rangle \mapsto -|\psi^-\rangle$$

Circuit operation on a standard basis measurement!

Example

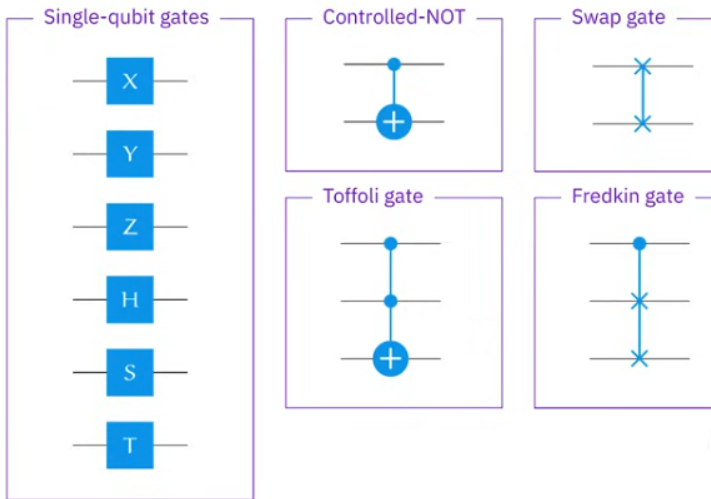


$$|\pi_0\rangle = |0\rangle|0\rangle$$

$$|\pi_1\rangle = |0\rangle|+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|01\rangle$$

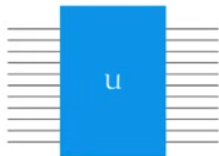
$$|\pi_2\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle = |\Phi^+\rangle$$

Qubits gates

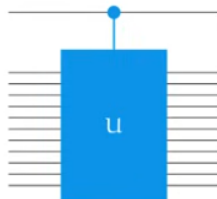


Unitary Operations

Unitary operation



Controlled-unitary operation



Inner Product

Suppose that we have two kets:

$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \quad \text{and} \quad |\phi\rangle = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix}$$

We then have

$$\langle\psi|\phi\rangle = (\overline{\alpha_1} \quad \cdots \quad \overline{\alpha_n}) \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix} = \overline{\alpha_1}\beta_1 + \cdots + \overline{\alpha_n}\beta_n$$

This is the *inner product* of $|\psi\rangle$ and $|\phi\rangle$.

Inner Product properties

Conjugate symmetry

For any two vectors

$$|\psi\rangle = \sum_{a \in \Sigma} \alpha_a |a\rangle \quad \text{and} \quad |\phi\rangle = \sum_{b \in \Sigma} \beta_b |b\rangle$$

we have

$$\langle \psi | \phi \rangle = \sum_{a \in \Sigma} \overline{\alpha_a} \beta_a \quad \text{and} \quad \langle \phi | \psi \rangle = \sum_{a \in \Sigma} \overline{\beta_a} \alpha_a$$

and therefore

$$\overline{\langle \psi | \phi \rangle} = \langle \phi | \psi \rangle$$

Inner Product properties

The Cauchy–Schwarz inequality

For every choice of vectors $|\psi\rangle$ and $|\phi\rangle$ we have

$$|\langle\psi|\phi\rangle| \leq \|\psi\| \|\phi\|$$

(Equality holds if and only if $|\psi\rangle$ and $|\phi\rangle$ are linearly dependent.)

Orthogonality and orthonormality

Two vectors $|\psi\rangle$ and $|\phi\rangle$ are **orthogonal** if their inner product is zero:

$$\langle\psi|\phi\rangle = 0$$

An **orthogonal set** $\{|\psi_1\rangle, \dots, |\psi_m\rangle\}$ is one where all pairs are orthogonal

$$\langle\psi_j|\psi_k\rangle = 0 \quad (\text{for all } j \neq k)$$

An **orthonormal set** $\{|\psi_1\rangle, \dots, |\psi_m\rangle\}$ is an orthogonal set of unit vectors:

$$\langle\psi_j|\psi_k\rangle = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases} \quad (\text{for all } j \neq k)$$

An **orthonormal basis** $\{|\psi_1\rangle, \dots, |\psi_m\rangle\}$ is an orthonormal set that forms a basis (of a given space).

Example

Example

For any classical state set Σ , the set of all standard basis vectors

$$\{|\alpha\rangle : \alpha \in \Sigma\}$$

is an orthonormal basis.

Example

The set $\{|+\rangle, |-\rangle\}$ is an orthonormal basis for the 2-dimensional space corresponding to a single qubit.

Example

The Bell basis $\{|\phi^+\rangle, |\phi^-\rangle, |\psi^+\rangle, |\psi^-\rangle\}$ is an orthonormal basis for the 4-dimensional space corresponding to two qubits.

Orthonormal bases connection with Unitary Matrices

Orthonormal bases are closely connected with unitary matrices.

These conditions on a square matrix U are equivalent:

1. The matrix U is unitary (i.e., $U^\dagger U = \mathbb{1} = U U^\dagger$).
2. The rows of U form an orthonormal basis.
3. The columns of U form an orthonormal basis.

For example, consider a 3×3 matrix U :

$$U^\dagger = \begin{pmatrix} \overline{\alpha_{1,1}} & \overline{\alpha_{2,1}} & \overline{\alpha_{3,1}} \\ \overline{\alpha_{1,2}} & \overline{\alpha_{2,2}} & \overline{\alpha_{3,2}} \\ \overline{\alpha_{1,3}} & \overline{\alpha_{2,3}} & \overline{\alpha_{3,3}} \end{pmatrix} \quad U = \begin{pmatrix} \alpha_{1,1} & \alpha_{1,2} & \alpha_{1,3} \\ \alpha_{2,1} & \alpha_{2,2} & \alpha_{2,3} \\ \alpha_{3,1} & \alpha_{3,2} & \alpha_{3,3} \end{pmatrix}$$

Example..

For example, consider a 3×3 matrix U :

$$U^\dagger = \begin{pmatrix} \overline{\alpha_{1,1}} & \overline{\alpha_{2,1}} & \overline{\alpha_{3,1}} \\ \overline{\alpha_{1,2}} & \overline{\alpha_{2,2}} & \overline{\alpha_{3,2}} \\ \overline{\alpha_{1,3}} & \overline{\alpha_{2,3}} & \overline{\alpha_{3,3}} \end{pmatrix} \quad U = \begin{pmatrix} \alpha_{1,1} & \alpha_{1,2} & \alpha_{1,3} \\ \alpha_{2,1} & \alpha_{2,2} & \alpha_{2,3} \\ \alpha_{3,1} & \alpha_{3,2} & \alpha_{3,3} \end{pmatrix}$$

Forming vectors from the columns of U , we can express $U^\dagger U$ like this:

$$|\psi_1\rangle = \begin{pmatrix} \alpha_{1,1} \\ \alpha_{2,1} \\ \alpha_{3,1} \end{pmatrix} \quad |\psi_2\rangle = \begin{pmatrix} \alpha_{1,2} \\ \alpha_{2,2} \\ \alpha_{3,2} \end{pmatrix} \quad |\psi_3\rangle = \begin{pmatrix} \alpha_{1,3} \\ \alpha_{2,3} \\ \alpha_{3,3} \end{pmatrix}$$

$$U^\dagger U = \begin{pmatrix} \langle \psi_1 | \psi_1 \rangle & \langle \psi_1 | \psi_2 \rangle & \langle \psi_1 | \psi_3 \rangle \\ \langle \psi_2 | \psi_1 \rangle & \langle \psi_2 | \psi_2 \rangle & \langle \psi_2 | \psi_3 \rangle \\ \langle \psi_3 | \psi_1 \rangle & \langle \psi_3 | \psi_2 \rangle & \langle \psi_3 | \psi_3 \rangle \end{pmatrix}$$

Projections

A square matrix Π is called a **projection** if it satisfies two properties:

1. $\Pi = \Pi^\dagger$
2. $\Pi^2 = \Pi$

Example

If $|\psi\rangle$ is a unit vector, then this matrix is a projection:

$$\Pi = |\psi\rangle\langle\psi|$$

$$\Pi^\dagger = (|\psi\rangle\langle\psi|)^\dagger = (\langle\psi|)^\dagger(|\psi\rangle)^\dagger = |\psi\rangle\langle\psi| = \Pi$$

$$\Pi^2 = (|\psi\rangle\langle\psi|)^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi| = \Pi$$

Summary: Key Takeaways I

- **Probabilistic Operations:** Transformations on compound systems use stochastic matrices acting on the Cartesian product of state spaces.
- **Matrix Tensor Products:** Joint operations are represented by the Kronecker product $(M \otimes N)$.
- **Quantum States:** Multiple qubits exist in the joint tensor product space $(\mathbb{C}^2)^{\otimes n}$.
- **State Vector Products:** Independent qubits combine as $|\phi\rangle \otimes |\psi\rangle$ following lexicographical ordering (e.g., $|00\rangle, |01\rangle, |10\rangle, |11\rangle$).
- **Unitary Multi-Systems:**
 - **Local:** $H \otimes I$ acts only on the first qubit.
 - **Global:** CNOT-like gates are inseparable and essential for entanglement.
- **Bilinearity:** Tensor products are linear in every "slot," allowing scalars to move: $\alpha(|A\rangle \otimes |B\rangle) = (\alpha|A\rangle) \otimes |B\rangle$.

Summary: Key Takeaways II

Next: Entanglement, Global Unitary Operations, and Multi-System Postulates.